

EAS 501 Lecture Notes 09/23/2026

Cheng Wang

Mathematics @ UMass Dartmouth

Outline

- Chapter 8. System of linear equations

8.2 Preliminaries 8.3 Gauss elimination

Linear system of equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = c_1, \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = c_2, \\ \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = c_m. \end{cases} \quad (1)$$

$$\text{Set } A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix}, \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \dots \\ c_m \end{pmatrix}.$$

Then linear system (1) could be rewritten as $A\mathbf{x} = \mathbf{c}$.

Augmented $m \times (n + 1)$ matrix

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & c_1 \\ a_{21} & a_{22} & \dots & a_{2n} & c_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & c_m \end{array} \right) \quad (2)$$

Elementary operations

- ▶ to multiply an equation by a non-zero scalar
- ▶ to add an equation multiplied
- ▶ to interchange two equations

Leading entry of a matrix: the first non-zero entry in a row.

Gauss elimination

To convert the augmented matrix into **row echelon form**

- ▶ leading entry shift to the right
- ▶ each leading entry is 1

$$\left(\begin{array}{cccccccccc|c} 1 & -1 & 3 & 0 & 2 & 1 & 4 & 0 & -3 & 7 & 2 \\ 0 & 1 & 1 & -2 & 0 & 0 & 6 & 1 & 3 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 3 & 1 & -4 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 9 & -1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Gauss Jordan reduction would lead to a **reduced row echelon form**, with leading entries as 1, and the other entries in its column are 0.

A few examples

Example 1
$$\begin{cases} x - y = 2, \\ 2x - y - z = 3, \\ x + y + z = 6. \end{cases}$$

$$\begin{aligned} (A|\mathbf{c}) &= \left(\begin{array}{ccc|c} 1 & -1 & 0 & 2 \\ 2 & -1 & -1 & 3 \\ 1 & 1 & 1 & 6 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & -1 & 0 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 2 & 1 & 4 \end{array} \right) \\ &\rightarrow \left(\begin{array}{ccc|c} 1 & -1 & 0 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right). \end{aligned}$$

It has been already in the row echelon and reduced row echelon form. An application of backward substitution gives the solution $x = 3$, $y = 1$, $z = 2$.

Example 2
$$\begin{cases} x + y - 2z = 1, \\ y - z = 3, \\ -x + 4y - 3z = 1. \end{cases}$$

$$\begin{aligned} (A|\mathbf{c}) &= \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 3 \\ -1 & 4 & -3 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 5 & -5 & 2 \end{array} \right) \\ &\rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & -13 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{array} \right). \end{aligned}$$

This is in the row echelon form, and the reduced row echelon

form is given by $\left(\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{array} \right)$. Since the last equation

is reduced to $0 \cdot x + 0 \cdot y + 0 \cdot z = 1$, this linear system is inconsistent.

Example 3
$$\begin{cases} x + y - 2z = 1, \\ y - z = 3, \\ -x + 4y - 3z = 14. \end{cases}$$

$$\begin{aligned} (A|\mathbf{c}) &= \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 3 \\ -1 & 4 & -3 & 14 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 5 & -5 & 15 \end{array} \right) \\ &\rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right). \end{aligned}$$

It has been already in the row echelon and reduced row echelon form. The reduced linear system takes a form of

$$\begin{cases} x - z = -2, \\ y - z = 3. \end{cases}$$
 There are infinitely many solutions, with a freedom of one-variable parameter.

Example 4
$$\begin{cases} x_1 + 2x_2 + 3x_3 + 4x_4 = 10, \\ x_2 + 2x_3 + 3x_4 = 6. \end{cases}$$

$$(A|\mathbf{c}) = \left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 10 \\ 0 & 1 & 2 & 3 & 6 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & -1 & -2 & -2 \\ 0 & 1 & 2 & 3 & 6 \end{array} \right).$$

It has been already in the row echelon and reduced row echelon form. The reduced linear system takes a form of

$$\begin{cases} x_1 - x_3 - 2x_4 = -2, \\ x_2 + 2x_3 + 3x_4 = 6, \end{cases} \text{ which in turn gives}$$

$$x_1 = x_3 + 2x_4 - 2, \quad x_2 = -2x_3 - 3x_4 + 6.$$

There are infinitely many solutions, with a freedom of two-variable parameter. By setting $x_3 = t$, $x_4 = s$, we get

$$\begin{cases} x_1 = t + 2s - 2, & x_2 = -2t - 3s + 6, \\ x_3 = t, & x_4 = s. \end{cases}$$